

Astronomy Question Bank

These questions were carefully written or selected to prepare students for the astronomy team selection process in Brazil. Some questions are not original; those are marked with () before the statement. The question-bank template is the same one used by Professor Kevin Zhou. His work is invaluable, and many ideas in this set can be found in his handouts.

1 Photometry and Modern Physics

- [3] **Problem 1.** The goal of this question is to derive an expression for the *optical depth*. Imagine that a beam of light passes through a region of space containing a certain number of particles. If I_0 is the intensity of the light before it passes through that region of space and I is the intensity afterward, the optical depth is defined as:

$$I = I_0 e^{-\tau}$$

where τ is the optical depth.

Consider a region of space with particle number density n .

- What is the number of particles in an area A and thickness dz ?
 - Assuming that each particle has cross section σ , what is the area blocked by the particles?
 - Find the formula for τ .
- [2] **Problem 2.** The Triangulum Galaxy, $M33$, is the third-largest galaxy in the Local Group. It lies at a distance $d = 970$ kpc from us and has an apparent magnitude of 5.72. Given that it contains approximately 40 billion stars, find the average luminosity of the stars in $M33$. Does your estimate seem consistent with reality? Why?
- [4] **Problem 3.** Consider a universe with a constant stellar number density, that is, a number of stars per unit volume, of value n . Assume that every star has $L = L_\odot$ and that the universe contains a total of N_0 stars.
- What is the probability that the absolute magnitude of a star, seen from the center of the universe, lies between m and $m + dm$, where dm is an infinitesimal interval of magnitude? Leave your answer in terms of m and the largest possible magnitude, m_{lim} , of a star at the "edge" of the universe.
 - What is the probability that a star can be observed with the naked eye?

Data:

$$\int_{-\infty}^6 10^{0.6(x-A)} dx \approx 2881.6 e^{-1.381A}$$

- [4] **Problem 4.** The *surface brightness*, flux per solid angle per frequency, B_ν , is given by Planck's law:

$$B_\nu = \frac{dF}{d\nu d\Omega} = \frac{2h\nu^3}{c^2(e^{\frac{h\nu}{k_B T}} - 1)}$$

- a) Find an expression for:

$$B_\lambda = \frac{dF}{d\lambda d\Omega}$$

- b) Starting from B_λ , find the peak wavelength λ_{max} emitted by a star of temperature T (you will have to solve something numerically).
- c) At low frequencies we have the Rayleigh-Jeans approximation. Obtain an expression for B_ν at low frequencies.

[3] **Problem 5.** The luminosity of a secondary body depends on the visible illuminated area of the object. In this question we briefly study that phenomenon.

- a) Consider the following situation:

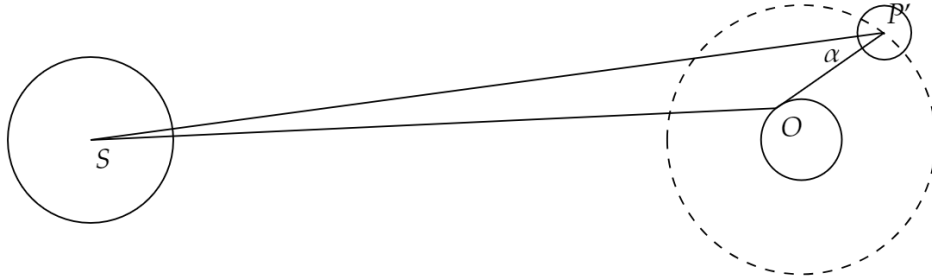


Figure 1: Sun-Earth-Moon diagram

Find an expression for the ratio Φ of the illuminated area, as a function of α , to the total area of the planet.

- b) Find the angles α at which the Moon is in the phases new, waxing, full, and waning, respectively.

[5] **Problem 6.** (Magna notes) In this problem we model the effect of the atmosphere on Earth. Suppose that the Sun is a blackbody of temperature T_1 and radius R_1 . Earth is a sphere located at a distance R from the Sun, with radius R_3 . Earth's emissivity is ϵ_3 .

- a) If Earth had no atmosphere, determine its equilibrium temperature, T_3 .
- b) Now we include the atmosphere. Model it as a spherical shell of gas, with emissivity ϵ_2 and outer radius $R_2 > R_3$, concentric with Earth. In thermal equilibrium, its absorptivity at ultraviolet and infrared wavelengths is ϵ_2 . The atmosphere transmits a fraction t of the ultraviolet radiation but is completely opaque to infrared radiation. Assuming that the Sun emits ultraviolet light while Earth emits and re-emits in the infrared, determine the temperature T_2 of the atmosphere and the temperature T_3 of Earth in thermodynamic equilibrium. Assume that the atmosphere is a perfect thermal conductor, so that all radiation incident on it is distributed uniformly over its surface.

- [2] **Problem 7.** (Problem set 2 - 2021) The Ring Nebula (M57) has an apparent magnitude of 9 and an angular diameter of $2'$ for an observer on Earth. What would be the apparent magnitude of the night sky of a planet orbiting a star exactly at the center of M57?
- [4] **Problem 8.** (Adapted from Problem set 4 - 2021) The Cherenkov effect was first detected by the Soviet scientist Pavel Cherenkov in 1937. Later, together with his colleagues I. E. Tamm and I. M. Frank, he gave the phenomenon a physical interpretation and was awarded the Nobel Prize in Physics in 1958. Before a mathematical treatment, we first need to understand the principle a little better.

When high-energy charged particles travel through a dielectric medium, atoms can be excited if the particle speed exceeds the phase velocity ($\frac{c}{n}$). Those atoms, in turn, emit electromagnetic radiation when they return to the ground state. The emitted waves spread spherically and, when superposed, form a cone of opening angle 2α , as the figure below shows.

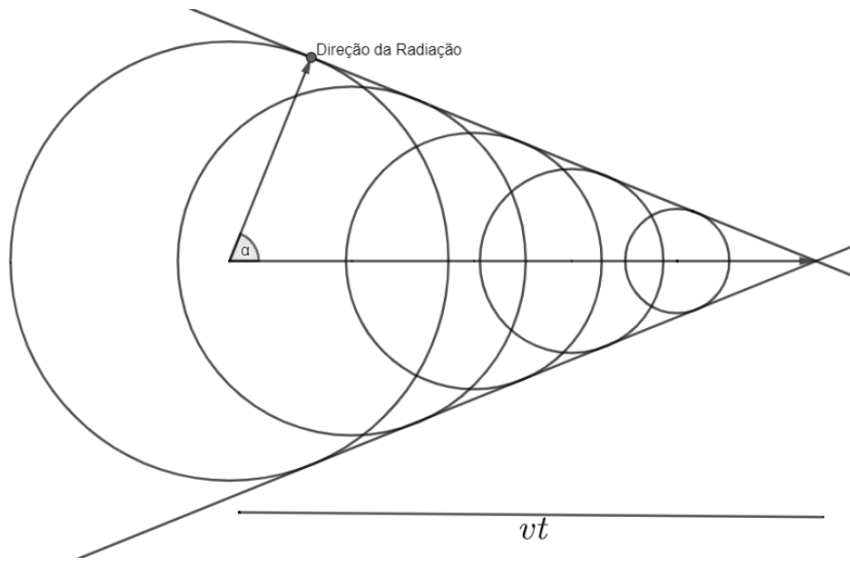


Figure 2: Radiation mechanism of the Cherenkov effect

This effect is similar to a jet moving at supersonic speed: it follows the same principle as the Mach cone, but with light. We now develop the mathematical model of the Cherenkov effect.

Part A - Theoretical Model

Consider a particle moving at relativistic speeds through a medium of refractive index n . Its rest mass is m_0 , and it has linear momentum p and speed v . At a given moment, a photon is emitted at an angle α , as Figure 1 shows.

- Letting μ be the frequency of the emitted photon, determine its linear momentum, p_μ , and its energy, E_μ . Your answer must be in terms of n , μ , and physical constants.
- Find an expression for the linear momentum of the particle after it emits the photon, in terms of p_μ , p , and α .

- c) Letting $\beta_n = \frac{c}{vn}$, prove that the relation below holds:

$$\cos \alpha = \frac{1}{\beta_n} \quad (1)$$

- d) Assuming that linear momentum and energy are conserved, determine the minimum speed at which the Cherenkov effect can occur. **Hint:** Compared with the other parameters, the factor $(n^2 - 1)h\mu$ may be neglected.

Part B - Nuclear Reactions

The proton-proton chain is a sequence of fusion reactions that convert hydrogen into helium. One possible branch of the proton-proton chain is *pp* IV, in which, in principle, a helium-3 atom reacts directly with a proton, according to the reaction below:



- e) State the nuclear conservation law you use, and identify the missing particle in the blank of the reaction above.
- f) State the nuclear conservation law you use, and identify the missing particle in the blank of the reaction below:

$$\pi^- \rightarrow \mu^- + ___ \quad (3)$$

Data: Pion mass: $140 \text{ MeV}/c^2$, muon mass: $106 \text{ MeV}/c^2$.

- [3] **Problem 9.** (Problem set 3 - Vinhedo 2022) Juvelino, observing directly from his observatory in Paris, France, monitors the star Polaris (αUMi). His goal is to find the color temperature T_c of the star. Some of the data he has on his target are:

- Apparent magnitude in the V band: $V = 1.98$;
- Absolute magnitude in the V band: $M_V = -3.60$;
- Absolute magnitude in the B band: $M_B = -3.19$.

Using the information provided, help Juvelino!

- a) From many observations, Juvelino determined that the interstellar extinction in the V band toward Polaris is $a_V = 5.8 \text{ mag/kpc}$. Find the distance, in pc, from αUMi to Earth.
- b) Using the empirical relation

$$\frac{A_V}{E_{B-V}} = 3.0 \quad (4)$$

where A_V is the total interstellar extinction in the V band and E_{B-V} is the $B - V$ color excess, find the $B - V$ color index of the observed star.

- c) Show the relation

$$T_c = \frac{7009}{(B - V) + 0.47} \quad (5)$$

in which the color temperature is given in kelvin. To do so, use the fact that stars of spectral class A0 have $(B - V) = 0$ and $T_c = 15000$ K. Also use the wavelengths of the B and V bands, $\lambda_B = 440$ nm and $\lambda_V = 548$ nm, respectively. Justify any approximations you make.

HINT: Planck's law may be useful

- d) Find the color temperature of Polaris.

[4] **Problem 10.** (Problem set 4 - 2020) Frightening as it is, Juvelino's banana plantation has a perfect sky for one of his two passions: astrophotography. He has a telescope 200 mm in diameter with a super CCD attached. Its parameters are: pixel side $5 \mu\text{m}$, plate scale 3.2 arcmin/mm, quantum efficiency 97%, RON (read-out noise) 1 count/pixel, and DN (thermal noise) 1 count/(pixel.hour). He prefers to observe in the V band (central wavelength $\lambda_V = 5500 \text{ \AA}$, bandwidth $\Delta\lambda = 820 \text{ \AA}$), because he remembers a magic number associated with it: the photon flux of a magnitude-0 object in this band is $\phi_0 = 1000 \text{ photons}/(\cdot \text{ cm}^2)$. Juvelino has two favorite targets: a star of magnitude $m_V = 4.51$ and a globular cluster of uniform surface brightness in V of $19.5 \text{ mag/arcsec}^2$ and angular diameter $\theta = 40''$.

- a) The night-sky brightness at Juvelino's farm is $21.9 \text{ mag/arcsec}^2$ in the V band, and the astronomical seeing is $\theta_0 = 1.2''$. Calculate the signal-to-noise ratio obtained in a photo of his favorite star with exposure time $t_{\text{exp}} = 30\text{s}$.

- b) Calculate the zero-point magnitude of Juvelino's telescope + CCD system, using the data for the star.

Juvelino's other great passion is Jiang, a Chinese university student who lives in Beijing. After years apart, Jiang suggests that Juvelino move to China. Before deciding, he looks up the night-sky brightness in the V band in his beloved's city, which is $19.3 \text{ mag/arcsec}^2$. That will affect imaging of the cluster, which is also visible from Jiang's home.

- c) Juvelino cannot live without photographing the cluster. He has inviolable rules for imaging this object, wherever he does it: reach a signal-to-noise ratio $S/N = 10$, and take the same number X of photos of the cluster in one night. X is the number of photos allowed by the sky conditions at his banana plantation. That number of frames per night is limited by the CCD battery (which lasts 3.2 hours) and by the exposure time required for each photo. Help him with this dilemma! Do the calculations, find how many photos of the cluster he can take in one night (in Beijing and at the farm), and conclude: will he see his beloved again, or will this love be too noisy?